# Mark Beers

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## *Education*

**University of California, Irvine** — September 2020 – Early 2026
*PhD - Cognitive Science* — *Irvine, California*
- Selected Coursework: Light and Geometry in Vision, Bayesian Cognitive Modeling, Optimization, Graph Algorithms, Deep Generative Models, Image Understanding

**University of California, Santa Cruz** — September 2017 – August 2019
*MS - Statistics* — *Santa Cruz, California*
- Selected Coursework: Bayesian Statistics, Probability, Spatial Statistics, Time Series

**University of California, Berkeley** — September 2012 – May 2017
*BS - Statistics, BA - Applied Mathematics* — *Berkeley, California*

## *Publications*

Beers, M. (2026). Perception of 3D mirror-symmetrical shapes from perspective images. *CURRENTLY UNDER REVIEW*.

Beers, M., & Pizlo, Z. (2024). Monocular reconstruction of shapes of natural objects. *Frontiers in Perception Science*, *18*. https://doi.org/10.3389/fnins.2024.1265966

VanDrunen, J., Nam, K., Beers, M., & Pizlo, Z. (2023). Traveling Salesperson Problem with Simple Obstacles: The Role of Multidimensional Scaling and the Role of Clustering. *Computational Brain and Behavior*, *6*(3), 513–525. https://doi.org/10.1007/s42113-022-00155-0

## *Experience*

**Teaching Assistant** — September 2020 – December 2025
*University of California, Irvine* — *Irvine, CA*
- Responsibilities typically involve leading discussion sections, grading homework and holding office hours
- I have performed this role for statistics, computer science and psychology courses offered within the School of Social Science at UCI.

**Instructor** — July 2023 – August 2023
*University of California, Irvine* — *Irvine, CA*
- Responsibilities: create lecture slides, homework and study material, lecture, hold office hours, write exams, assign work to the teaching assistant
- Course Content: Hypothesis testing, ANOVAs, Linear Regression

**Teaching Assistant** — September 2018 – August 2019
*University of California, Santa Cruz* — *Santa Cruz, CA*
- Responsibilities typically involve leading discussion sections, grading homework and holding office hours
- I assisted with courses taught by the statistics and mathematics departments.

**Intern, Renewable and Appropriate Energy Laboratory** — Fall 2016 – Spring 2017
*University of California, Berkeley* — *Berkeley, CA*

- I assisted in the application of the SWITCH model to Costa Rica. The SWITCH model was developed to help identify the optimal set of power plants to add to the electricity grid in a certain geographic area. Specifically, I wrote Python scripts to acquire, clean and upload data to an SQL database referenced by the SWITCH model. I also performed data visualization and analysis in R.

**Intern** Summer 2016, Summer 2017

*Astro Aerospace* *Carpinteria, CA*

- I wrote Python and Matlab scripts involved with analyzing the RF performance of antennas, developed a tutorial for common operations in TICRA's reflector analysis program, GRASP and generally assisted in the mechanical design and analysis of antennas.

## *Conference Presentations*

Beers, M. (2025). A bayesian hierarchical model for recovering 3d natural shapes from perspective images. *Vision Sciences Society*. https://jov.arvojournals.org/article.aspx?articleid=2809988

Beers, M., & Pizlo, Z. (2024). The role of uncertain perspective information in recovering 3d symmetrical shapes. *Vision Sciences Society*. https://jov.arvojournals.org/article.aspx?articleid=2801099

Beers, M., & Pizlo, Z. (2023). The role of orthogonality and compactness in recovering 3d symmetrical shapes. *Vision Sciences Society*. https://jov.arvojournals.org/article.aspx?articleid=2791669&resultClick=1

Beers, M., & Pizlo, Z. (2022). Monocular 3d reconstruction of polyhedral shapes via neural network. *MODVIS*. https://docs.lib.purdue.edu/modvis/2022/session02/3/

## *Specialized Skills*

**Programming Languages**: Python, R, Javascript, HTML, CSS, SQL
**Frameworks**: PyTorch, JAX
**Other Skills**: Psychophysics, Experiment Design, Bayesian Statistics, Optimization (Nonconvex, Mixed Integer Linear Programming, Multi-objective,...)

## *Interests*

**Professional Interests**: Human Vision, Computer Vision, Bayesian Statistics, Optimization, Public Policy Statistics, Explainable AI, Time Series Analysis
**Athletics**: Mountain Biking, Climbing, Snowboarding